Task 2: Electricity Demand Forecasting

InternSpark Business Analytics Internship | PJME hourly electricity consumption

Executive summary

This project uses the supplied PJME hourly electricity-consumption series, covering 2002-01-01 to 2018-08-03. After timestamp cleaning and aggregation, 6,059 daily average-demand observations were prepared. Mean daily average demand was 32,080 MW; the observed daily-average range was 19,769 MW (2012-10-30) to 52,230 MW (2006-08-02). An additive Holt–Winters ETS model with weekly seasonality was evaluated on the final 365 days. It achieved MAE 5,577 MW and MAPE 19.23%; the weekly seasonal-naive benchmark achieved MAE 4,839 MW and MAPE 16.18%. The 90-day base forecast averages 34,702 MW. These are historical-model results, not a current operational forecast because the dataset ends in 2018.

Data preparation and assumptions

The raw file contained 145,366 usable rows. 8 rows were part of duplicated timestamps; readings at the same timestamp were averaged. Reindexing the series to hourly frequency identified 30 missing hourly timestamps. Daily mean demand was calculated and 0 missing daily values were interpolated to create a continuous series. This creates a regular index but introduces estimated values where observations were absent. The assignment dataset measures electricity demand in MW, so the project interprets the requested 'sales forecasting' task as demand forecasting.

Model evaluation

Model	MAE (MW)	MAPE (%)	RMSE (MW)
ETS (additive trend + weekly seasonality)	5,577.3	19.23	6,321.8
Weekly seasonal-naive benchmark	4,838.8	16.18	5,781.5

Forecast and uncertainty

The forecast covers 2018-08-04 through 2018-11-01. The first-day point forecast is 33,062 MW, and the final-day point forecast is 35,741 MW. Approximate 90% prediction intervals are generated by resampling one-year holdout residuals. The low and high scenarios apply -5% and +5% to the base forecast as illustrative sensitivities; they are not confidence limits.

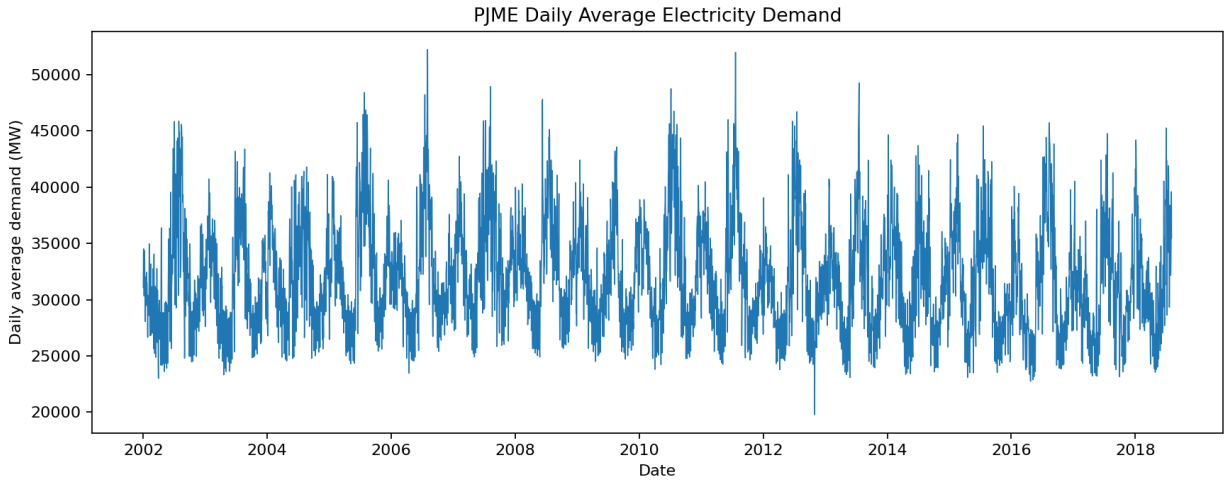


Figure 1. Daily average demand across the historical series.

Forecast diagnostics and business actions

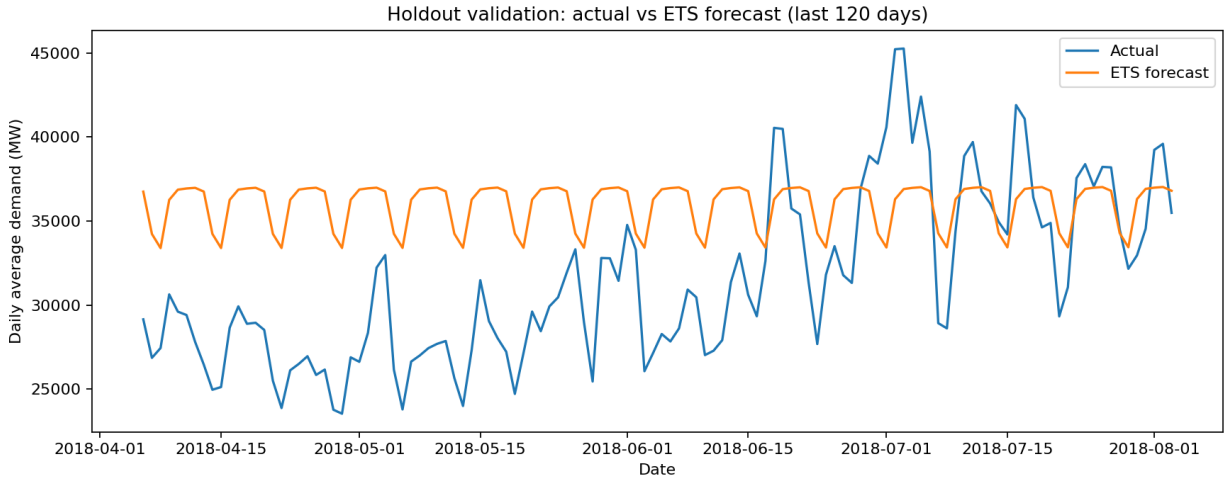


Figure 2. Actual demand compared with ETS predictions over the final 120 days of the holdout period.

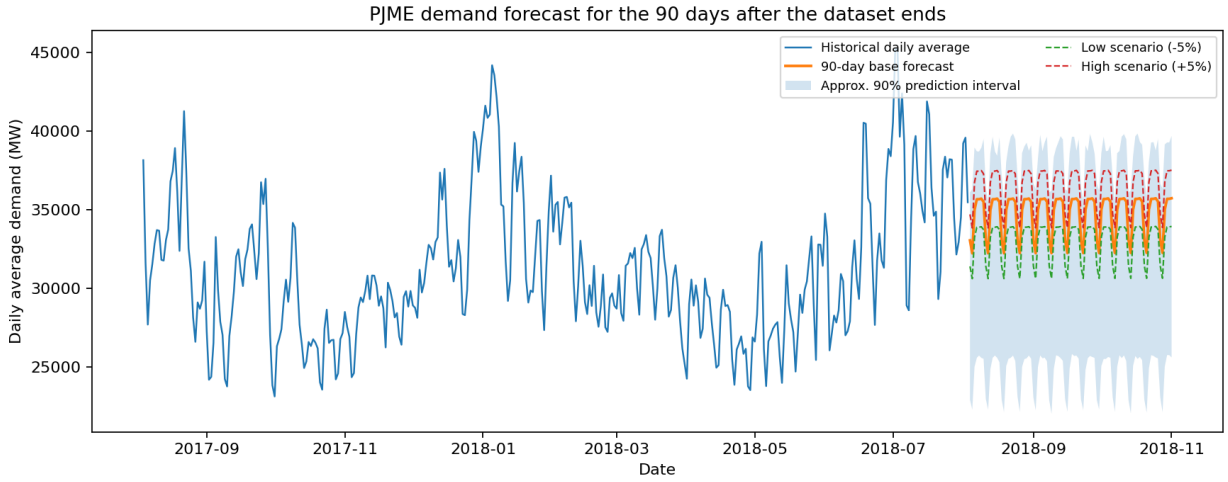


Figure 3. Ninety-day forecast, approximate interval, and illustrative scenarios.

Recommended business actions

• **Capacity planning:** use the base forecast as a planning input and check reserve capacity against the upper interval.

• **Scenario readiness:** test operational plans under the ±5% demand sensitivities; do not treat these scenarios as probabilities.

• **Forecast governance:** track daily MAE/MAPE after each new observation and retrain if errors persistently rise.

• **Model improvement:** evaluate temperature, holidays, and calendar effects in a future version; these drivers are not included in this univariate model.

• **Data quality:** maintain a missing-timestamp log and distinguish measured values from interpolated values in operational reporting.

Limitations

The data is historical and ends in August 2018. ETS models weekly seasonality but does not explicitly model annual weather patterns, holidays, or structural changes. The residual-bootstrap interval depends on holdout error patterns and is approximate. The ±5% scenario bounds are assumptions. The model should be retrained and validated on recent data before operational use.